

MINOS-Bench: Supplementary Materials

Anonymous Submission

Contents

1	Semiconductor Domain Knowledge Base	3
1.1	Manufacturing Processes (60 terms, weight = 1.5)	3
1.2	Materials Science (51 terms, weight = 1.3)	3
1.3	Device Physics (62 terms, weight = 1.4)	3
1.4	Process Nodes (65 terms, weight = 1.6)	4
1.5	Testing Methodologies (51 terms, weight = 1.2)	4
1.6	Applications (62 terms, weight = 1.1)	4
1.7	Performance Parameters (80 terms, weight = 1.7)	4
1.8	Knowledge Base Design Rationale	5
2	Generation System Implementation Details	6
2.1	Content Analysis and Precision Classification	6
2.2	Parameter Selection Rationale	6
2.3	Adaptive Temperature Control	6
2.4	Adaptive Similarity Thresholds	8
2.5	Cross-Document Linking Parameters	8
2.6	Document Chunking	8
3	Dynamic Prompt Templates	9
3.1	Base Prompt Template	9
3.2	Test-Type Specific Instructions	9
3.2.1	Robustness Testing	9
3.2.2	Multi-Hop Reasoning	10
3.2.3	Generation Quality	10
3.3	Question Format Specifications	10
3.3.1	Format Balance Control	10
3.3.2	Objective Question Types	11
3.3.3	Subjective Question Archetypes	12
3.4	Diversity Enhancement Instructions	13
3.5	Complete Prompt Construction Example	13
3.6	Prompt Component Interaction	13
4	Evaluation Rubrics	15
4.1	Completeness	15
4.1.1	Objective Questions	15
4.1.2	Subjective Questions	15
4.2	Depth	15

4.2.1	Objective Questions	15
4.2.2	Subjective Questions	16
4.3	Factuality	16
4.3.1	With Retrieval Context	16
4.3.2	Without Retrieval Context	17
4.4	Relevance	18
4.4.1	Objective Questions	18
4.4.2	Subjective Questions	18
4.5	Context Utilization	19
4.5.1	With Retrieval Context	19
4.5.2	Manual Setup Frameworks	19
4.6	Support Quality	20
4.6.1	Objective Questions	20
4.6.2	Subjective Questions	20
4.7	Implementation Details	21
5	Evaluation Platform Implementation	22
5.1	Framework Capability Matrix	22
5.2	Adaptive Evaluation Strategy	22
5.3	Dual-Mode Implementation	23
5.4	Output Format	23
6	Complete Experimental Results	24

1 Semiconductor Domain Knowledge Base

This appendix provides the complete listing of 431 technical terms in our semiconductor domain knowledge base, organized into seven categories with associated precision weights.

1.1 Manufacturing Processes (60 terms, weight = 1.5)

Core fabrication techniques: wafer, lithography, etching, CVD, PVD, ALD, ion implantation, annealing, CMP, cleaning, inspection, metrology, mask, exposure, develop, strip, deposition, diffusion, oxidation, epitaxy, packaging, wire bonding, flip chip, WLP, SiP, BGA, CSP

Advanced lithography: immersion lithography, multi-patterning, self-aligned, spacer, hard mask, anti-reflective coating, photoresist, reticle

Etching techniques: reactive ion etching, RIE, plasma etching, dry etching, wet etching, isotropic etching, anisotropic etching

Deposition methods: PECVD, LPCVD, MOCVD, MBE, electroplating, sputtering, atomic layer deposition, physical vapor deposition

CMP and cleaning: chemical mechanical planarization, post-CMP cleaning, megasonic cleaning, RCA clean, HF dip

Metrology: overlay metrology, critical dimension measurement, ellipsometry, optical CD, scatterometry

1.2 Materials Science (51 terms, weight = 1.3)

Substrate materials: silicon, GaAs, GaN, SiC, InP, germanium, polysilicon, monocrystalline, amorphous

Dielectric materials: high-k, low-k, SiO₂, Si₃N₄, Al₂O₃, HfO₂, silicon dioxide, silicon nitride, hafnium oxide, aluminum oxide, zirconium oxide, tantalum pentoxide, ultra-low-k, porous silica, carbon-doped oxide

Metal materials: copper, aluminum, tungsten, titanium, tantalum, indium, silver, gold, platinum, nickel, cobalt, ruthenium, molybdenum

Compound semiconductors: gallium arsenide, gallium nitride, silicon carbide, indium phosphide, aluminum nitride, zinc oxide

2D materials: graphene, MoS₂, transition metal dichalcogenides

Gate stack materials: metal gate, work function metal, barrier layer, capping layer, etch stop layer

1.3 Device Physics (62 terms, weight = 1.4)

Transistor types: MOSFET, FinFET, GAAFET, CMOS, BJT, JFET, IGBT, diode, transistor

Advanced transistors: gate-all-around, nanosheet, nanowire, CFET, complementary FET, vertical transistor, TFET, tunnel FET, FDSOI, fully depleted SOI

Memory devices: memory, DRAM, SRAM, NAND, NOR, FRAM, ReRAM, MRAM, flash memory, 3D NAND, PCM, phase change memory, RRAM, STT-MRAM, emerging memory

Passive components: resistor, capacitor, inductor

Integrated circuits: IC, integrated circuit, ASIC, SoC, system on chip, logic circuit, analog circuit, mixed-signal

Sensors and MEMS: MEMS, sensor, actuator, image sensor, CMOS sensor, pressure sensor, accelerometer, gyroscope

Specialty devices: RF, power device, optoelectronic, LED, laser diode, photodiode, solar cell, TFT, thin film transistor

1.4 Process Nodes (65 terms, weight = 1.6)

Process nodes: 14nm, 10nm, 7nm, 5nm, 3nm, 2nm, 1nm, 28nm, 22nm, 16nm, 12nm, 8nm, 6nm, 4nm, process node, technology node

Lithography technologies: EUV, DUV, 193nm, 248nm, 365nm, ArF, KrF, i-line, g-line, extreme ultraviolet, deep ultraviolet, 13.5nm, high-NA EUV

Process integration: FinFET process, SOI, FD-SOI, bulk, bulk silicon, strained silicon, strain engineering, SiGe, silicon germanium, HEMT, high electron mobility

Process stages: FEOL, BEOL, MEOL, front end of line, back end of line, middle of line

Advanced techniques: gate-first, gate-last, replacement gate, high-k metal gate, HKMG, multiple patterning, SADP, SAQP, self-aligned double patterning, self-aligned quadruple patterning, LELE, LFLE

Interconnect: damascene, dual damascene, copper interconnect, via, through silicon via, TSV, backside power delivery

1.5 Testing Methodologies (51 terms, weight = 1.2)

Reliability testing: reliability, yield, failure analysis, electrical test, functional test, burn-in, temperature cycling, thermal shock, humidity test

Failure mechanisms: electromigration, hot carrier, NBTI, TDDB, negative bias temperature instability, time dependent dielectric breakdown, stress migration, void formation

Quality control: SPC, statistical process control, parameter drift, defect density, process capability, Cpk, design of experiments, DOE

Inspection techniques: critical dimension, CD-SEM, AFM, SEM, TEM, X-ray, atomic force microscopy, scanning electron microscopy, transmission electron microscopy, optical inspection, e-beam inspection

Electrical characterization: I-V curve, C-V measurement, DLTS, deep level transient spectroscopy, Hall effect, mobility measurement, sheet resistance, four-point probe, contact resistance

Wafer-level testing: probe test, wafer sort, parametric test, functional test, final test, package test

1.6 Applications (62 terms, weight = 1.1)

Computing processors: AI chip, GPU, CPU, NPU, TPU, FPGA, ASIC, graphics processing unit, neural processing unit, tensor processing unit, application specific IC, DSP, digital signal processor, MCU, microcontroller

Computing systems: edge computing, cloud computing, data center, server, supercomputer, quantum computing, neuromorphic computing, in-memory computing

Consumer electronics: smartphone, tablet, laptop, wearable, smartwatch, AR glasses, VR headset

Automotive: automotive electronics, ADAS, autonomous driving, LiDAR, radar, infotainment, powertrain

Communication: 5G, 6G, IoT, internet of things, wireless, RF transceiver, baseband, modem

AI/ML: neural network, machine learning, deep learning, inference, training, transformer, large language model

Emerging applications: blockchain, cryptocurrency, mining, virtual reality, VR, augmented reality, AR, metaverse, HPC, high performance computing

1.7 Performance Parameters (80 terms, weight = 1.7)

Voltage parameters: threshold voltage, V_{th} , supply voltage, VDD, VSS, voltage, operating voltage, breakdown voltage, junction temperature

Current parameters: leakage current, drive current, I_{on} , I_{off} , current, saturation current, subthreshold current, gate leakage, junction leakage

Power parameters: power consumption, static power, dynamic power, power dissipation, TDP, thermal design power, power density, energy efficiency

Timing parameters: switching speed, delay, propagation delay, rise time, fall time, setup time, hold time, clock skew, clock frequency, access time

Frequency parameters: frequency, bandwidth, cutoff frequency, f_T , f_{max} , operating frequency, clock speed

Resistance and capacitance: resistance, capacitance, R_{on} , on-resistance, gate capacitance, parasitic capacitance, RC delay, interconnect resistance

Temperature parameters: operating temperature, junction temperature, thermal resistance, temperature coefficient

Noise and linearity: noise, noise figure, SNR, signal to noise ratio, linearity, THD, total harmonic distortion

RF parameters: gain, phase, S-parameters, impedance matching, insertion loss, return loss

Performance metrics: DIBL, drain induced barrier lowering, subthreshold swing, SS, transconductance, g_m , output conductance, GIDL, gate induced drain leakage, mobility, carrier mobility, saturation velocity

1.8 Knowledge Base Design Rationale

The knowledge base was designed with several key considerations:

Coverage breadth: The 431 terms span the full semiconductor technology stack from materials science to system applications, ensuring comprehensive domain coverage for question generation across all technical sophistication levels.

Precision weighting: Category weights ($w_i \in [1.1, 1.7]$) reflect the technical rigor requirements, with parameters (1.7) and processes (1.6) weighted highest due to their quantitative precision requirements, while applications (1.1) receive lower weights as they involve more qualitative discussion.

Term granularity: The knowledge base includes both high-level concepts (e.g., “transistor”) and specific implementations (e.g., “FinFET”, “GAA”, “nanosheet”) to enable accurate classification of technical content at varying abstraction levels.

Temporal coverage: Terms encompass mature technologies (e.g., “14nm”, “planar MOSFET”), current leading-edge (e.g., “3nm”, “GAA”), and emerging concepts (e.g., “CFET”, “backside power”), ensuring the benchmark remains relevant as technology advances.

This comprehensive knowledge base enables the generation system to accurately measure technical content density ($\rho(d)$) and assign appropriate precision levels ($p(d)$), critical for adaptive question generation as described in Section ??.

2 Generation System Implementation Details

This appendix provides specific parameter values for the adaptive benchmark generation system described in Section ???. These values were empirically determined through iterative experimentation on held-out document sets.

2.1 Content Analysis and Precision Classification

Technical Density. Technical density $\rho(d)$ measures domain term concentration (Eq. ?? in main text). Documents are classified into precision levels based on density and weighted high-precision category presence:

$$p(d) = \begin{cases} \text{high} & \text{if } \rho(d) > 0.20 \text{ or } \omega_h(d) > 8 \\ \text{medium} & \text{if } \rho(d) > 0.12 \text{ or } \omega_h(d) > 4 \\ \text{low} & \text{otherwise} \end{cases} \quad (1)$$

where $\omega_h(d) = \sum_{c_i \in \{\text{parameters, processes}\}} w_i \cdot |\{t \in T_{c_i} : n(t, d) > 0\}|$ represents weighted count of high-precision category terms.

The disjunctive logic ensures documents containing critical parameters (e.g., specific voltage values, process nodes) are classified as high-precision even with low overall density.

2.2 Parameter Selection Rationale

The parameter values presented in this appendix were determined through a combination of established guidelines from the literature and empirical tuning on held-out document sets.

For temperature control, we follow the foundational work of Holtzman2020The, which establishes that: (1) lower temperatures produce more deterministic outputs suitable for precision-critical tasks, (2) higher temperatures increase diversity but risk incoherence, and (3) nucleus sampling with adaptive top- p helps maintain quality during high-temperature exploration. Recent empirical studies [?] confirm that optimal temperature varies by task type, supporting our content-aware adaptive strategy.

The specific threshold values (e.g., precision classification at $\rho > 0.20$, similarity thresholds of 0.70–0.80) were calibrated through grid search on 50 representative documents from our corpus, selecting values that balanced generation success rate with output diversity. These values may require adjustment for corpora with different technical density distributions.

2.3 Adaptive Temperature Control

Base Temperature Ranges. Temperature ranges vary by precision level to balance accuracy with diversity:

$$[\tau_{\min}(p), \tau_{\max}(p)] = \begin{cases} [0.4, 0.8] & p = \text{high} \\ [0.5, 0.9] & p = \text{medium} \\ [0.6, 1.0] & p = \text{low} \end{cases} \quad (2)$$

Progressive Temperature. Base temperature progresses with successful generations to encourage diversity:

$$\tau_{\text{progress}}(k) = (\tau_{\max} - \tau_{\min}) \times \min\left(\frac{k}{20}, 0.8\right) \quad (3)$$

where k is the count of successfully generated questions. This allows early questions to prioritize precision while later questions explore diverse formulations.

Category-Specific Adjustments. Fine-grained temperature adjustments based on dominant content category:

$$\Delta\tau_c(c^*) = \begin{cases} -0.10 & c^* = \text{parameters} \\ -0.08 & c^* = \text{processes} \\ -0.05 & c^* = \text{devices} \\ -0.03 & c^* = \text{testing} \\ 0.00 & c^* = \text{materials} \\ +0.02 & c^* = \text{manufacturing} \\ +0.05 & c^* = \text{applications} \end{cases} \quad (4)$$

Categories requiring higher precision (parameters, processes) receive negative adjustments to lower temperature, while categories permitting more flexibility (applications) receive positive adjustments.

Failure Recovery Boosts. Temperature increases after generation failures to escape repetitive patterns:

$$\Delta\tau_{\text{attempt}}(a) = \min(0.25, 0.08 \times a) \quad (5)$$

$$\Delta\tau_{\text{similarity}}(s) = \min(0.15, 0.05 \times s) \quad (6)$$

where a counts all failed attempts and s tracks consecutive similarity failures. Similarity failures receive cumulative boosts from both terms ($\Delta\tau_{\text{attempt}} + \Delta\tau_{\text{similarity}}$), providing stronger recovery for diversity issues than format failures alone.

Final Temperature Computation.

$$\tau = \text{clip}\left(\tau_{\min} + \tau_{\text{progress}} + \Delta\tau_c + \Delta\tau_{\text{attempt}} + \Delta\tau_{\text{similarity}}, 0.1, 1.0\right) \quad (7)$$

Complementary Nucleus Sampling. To maintain coherence during high-temperature exploration, nucleus sampling uses adaptive top- p :

$$p_{\text{nucleus}}(\tau) = \begin{cases} 0.95 & \tau \leq 0.4 \\ 0.90 & 0.4 < \tau \leq 0.7 \\ 0.85 & \tau > 0.7 \end{cases} \quad (8)$$

Lower p at higher temperatures prevents sampling from low-probability token tails.

Example Evolution. For a medium-precision process document ($p = \text{medium}$, $\rho = 0.15$, $c^* = \text{processes}$):

- **Initial** ($k = 0$): $\tau = 0.5 - 0.08 = 0.42$ (controlled precision)
- **After 10 successes** ($k = 10$): $\tau = 0.5 + 0.4 \times 0.5 - 0.08 = 0.62$ (exploring)
- **After 3 failures** ($a = 3, s = 2$): $\tau = 0.62 + 0.24 + 0.10 = 0.96$ (diversity boost)

2.4 Adaptive Similarity Thresholds

Base Thresholds by Precision Level.

$$\theta_{\text{base}}(p) = \begin{cases} 0.70 & p = \text{high} \\ 0.75 & p = \text{medium} \\ 0.80 & p = \text{low} \end{cases} \quad (9)$$

Lower thresholds for high-precision content acknowledge that questions about specialized topics (e.g., ALE processes, FinFET structures) necessarily share technical terminology while remaining substantively different.

Progressive Relaxation. Threshold decreases with retry count to enable graceful degradation:

$$\theta_{\text{sim}}(r) = \max(0.50, \theta_{\text{base}}(p) - 0.05 \times r) \quad (10)$$

The floor of 0.50 prevents acceptance of near-duplicate questions even after multiple failures.

Weighted Similarity Computation. Two-stage uniqueness validation:

$$\text{sim}_{\text{weighted}}(q, q') = \text{Jaccard}(W_q, W_{q'}) + \alpha \cdot \text{Jaccard}(T_q, T_{q'}) \quad (11)$$

where W denotes all word tokens, $T \subseteq W$ denotes technical terms, and $\alpha = 0.05$ is the technical term penalty weight. The low weight on technical term overlap allows questions sharing domain vocabulary to be considered unique if they differ in non-technical aspects.

For corpora with ≥ 5 existing questions, TF-IDF semantic similarity provides additional validation:

$$\text{sim}_{\text{semantic}}(q, q') = \cos(\vec{v}_{\text{TF-IDF}}(q), \vec{v}_{\text{TF-IDF}}(q')) \quad (12)$$

Questions are rejected if $\max_{q' \in \mathcal{Q}} \text{sim}_{\text{semantic}}(q, q') > \theta_{\text{sim}} + 0.05$, with the relaxed threshold preventing false rejections.

2.5 Cross-Document Linking Parameters

The semantic similarity threshold for cross-document linking is $\theta_{\text{link}} = 0.1$, serving as a minimum relatedness requirement. This relatively low threshold ensures meaningful connections between documents discussing related concepts from different perspectives (e.g., a patent’s process parameters linked to a paper’s theoretical analysis).

2.6 Document Chunking

Documents are segmented using sliding window approach:

- Chunk size: 512 tokens
- Overlap: 128 tokens (25%)

This configuration preserves semantic coherence within chunks while enabling fine-grained context construction for cross-document synthesis. S

3 Dynamic Prompt Templates

This appendix provides the complete prompt templates used in our dynamic prompt engineering system (Section ??), demonstrating how different components combine to create scenario-specific, format-aware, and diversity-conscious generation instructions.

3.1 Base Prompt Template

All prompts begin with a common base structure defining the role, context, and output format:

Base Template Structure

ROLE & GOAL

You are a world-class AI Benchmark Scientist at a leading semiconductor research institute. Your mission is to create ONE exemplary benchmark use case from the provided technical documentation. This use case is specifically designed to test an AI assistant’s capabilities based on a predefined TEST TYPE and SCENARIO.

CONTEXT

Here is the context, composed of several sections from a technical standard. Analyze it for potential complexities and inter-dependencies.

[Context sections inserted here]

SCENARIO-SPECIFIC INSTRUCTIONS

TEST TYPE: {test_type}

DESCRIPTION: {test_description}

INSTRUCTIONS: {specific_instructions}

OUTPUT FORMAT

You must respond with ONLY a valid JSON object in the following format. Do not include any text before or after the JSON.

3.2 Test-Type Specific Instructions

3.2.1 Robustness Testing

ROBUSTNESS CONFIGURATION

Test Description: This test case evaluates the RAG system’s robustness in retrieving a specific, rare, or hard-to-find fact (the “needle”) from a large context (“haystack”).

Specific Instructions:

1. Select a single, highly specific fact or detail from the CONTEXT that is only mentioned once or is easily overlooked
2. Formulate a question that requires the model to find and accurately reproduce this “needle” fact
3. Construct a ground truth answer that is concise, unambiguous, and directly quotes or paraphrases the exact fact from the context

Primary Metric: Factuality

Secondary Metrics: Depth, Completeness, Relevance, Context Utilization, Support Quality

3.2.2 Multi-Hop Reasoning

MULTI-HOP CONFIGURATION

Test Description: This test case evaluates the system’s ability to retrieve and synthesize information from multiple, separate document sections to answer a complex question.

Specific Instructions:

1. Identify a logical chain of information that requires at least two separate sections from the CONTEXT to connect
2. Formulate a question that forces this multi-hop reasoning process
3. Construct a ground truth answer that explicitly synthesizes information from these different sources

Primary Metric: Completeness

Secondary Metrics: Depth, Factuality, Relevance, Context Utilization, Support Quality

3.2.3 Generation Quality

GENERATION QUALITY CONFIGURATION

Test Description: This test case evaluates the generator model’s accuracy and technical depth, assuming a perfect retrieval process.

Specific Instructions:

1. Assume the user has perfect context. Formulate a question that requires deep reasoning (analysis, comparison, explanation of cause-and-effect) rather than simple fact extraction
2. Construct a ground truth answer that demonstrates profound understanding and insight based on the context

Primary Metric: Depth

Secondary Metrics: Completeness, Factuality, Relevance, Context Utilization, Support Quality

3.3 Question Format Specifications

3.3.1 Format Balance Control

To ensure comprehensive benchmark coverage, we maintain target distribution of 40% objective and 60% subjective questions. The system actively tracks the objective question ratio during generation:

$$r_{\text{obj}} = \frac{n_{\text{obj}}}{n_{\text{obj}} + n_{\text{subj}}} \quad (13)$$

When $r_{\text{obj}} < 0.5$, the system forces objective question generation to restore balance. Among objective questions, target ratios are maintained through balanced selection: Mathematical calculation (30%), Fill-in-blank (25%), True/False (25%), Multiple choice (20%).

3.3.2 Objective Question Types

1. Mathematical Calculation (Target: 30%). **Format:** [Calculation] Question text requiring numerical computation

Example: Based on the transistor parameters provided ($V_{DD}=3.3V$, $C_L=15fF$, $I_{on}=80\mu A$), calculate the propagation delay using the specific formula mentioned in the context. Show all steps and use exact parameter values.

Requirements:

- MUST extract ALL numerical parameters from context
- MUST use context-specific formulas, NOT generic ones
- MUST show step-by-step calculation with units
- MUST identify if context provides special conditions or adjustments
- Trap: Generic formulas without context adjustment should fail

2. Fill-in-Blank (Target: 25%). **Format:** [Fill] Statement with _____ blank

Example: According to the patent specification, the exact threshold voltage for the 28nm FinFET process with high-k dielectric at junction temperature $125^{\circ}C$ is _____ V

Requirements:

- Extract EXACT numerical values with specified conditions
- Must include units and precision as stated in context
- Trap: Approximate or generic values should be wrong

3. True/False (Target: 25%). **Format:** [Judge] Technical statement (True/False)

Example: Based on the specific process described, the external epitaxial layer is completely consumed during silicidation while the shallow implant region covers the silicide area, preventing leakage. (True/False)

Requirements:

- Create statements about SPECIFIC process mechanisms
- MUST require understanding of multi-step processes
- CREATE BOTH true AND false statements (not always true)
- Trap: General knowledge insufficient—only context-specific details give correct answer

4. Multiple Choice (Target: 20%). Format: [Choice] Question? A) Option1 B) Option2 C) Option3 D) Option4

Example: What is the PRIMARY reason the described CMP process uses IM1/IM2 selectivity over IL2? A) Cost reduction B) 10:1 polish rate difference enables precise endpoint detection C) Temperature control D) Chemical compatibility

Requirements:

- Format: All A) B) C) D) options in ONE line
- Question WILL BE REJECTED if options are missing
- Create options where ONLY context provides distinguishing technical details
- Each option should be 1–2 lines maximum
- DO NOT write “which of the following” without actual options

3.3.3 Subjective Question Archetypes

SUBJECTIVE FORMAT ROTATION

For subjective questions (target: 60% of benchmark), the system cycles through five question archetypes to ensure stylistic diversity. Current archetype is determined by: $(\text{question_count}) \bmod 5$.

Five Question Archetypes:

1. **Definition/Specification:** “Ask for a specific definition or specification value”

Example: What is the exact definition of “effective channel length” as specified in the provided semiconductor process documentation, and what are the key factors that distinguish it from the drawn channel length?

2. **Process Explanation:** “Ask to explain a process or sequence of events”

Example: Describe the complete sequence of steps involved in the dual damascene process as outlined in the context, explaining how each step contributes to the final metal interconnect structure.

3. **Causal Reasoning:** “Ask for the reason or purpose behind a specification (a ‘why’ question)”

Example: Why does the described FinFET structure use high-k metal gate instead of traditional poly-silicon gate, and what specific performance improvements does this enable according to the context?

4. **Comparative Analysis:** “Ask to compare or contrast two related terms or concepts found in the text”

Example: Compare the trade-offs between arsenic and phosphorus doping for source/drain implantation as discussed in the context, considering activation temperature, diffusion characteristics, and junction depth control.

5. **Problem Identification:** “Ask about a potential problem, defect, or limitation mentioned”

Example: What are the primary reliability concerns associated with copper interconnects discussed in the documentation, and what mitigation strategies does the process flow incorporate to address these issues?

3.4 Diversity Enhancement Instructions

When similarity check failures exceed threshold ($s > 2$), the system injects a concise diversity directive into prompts. The directive is intentionally lightweight, providing just enough guidance to encourage exploration without over-constraining generation:

DIVERSITY DIRECTIVE STRUCTURE

DIVERSITY FOCUS: Previous s uniqueness attempts suggest trying a new approach

RECOMMENDED STYLE: Focus on “[current archetype from rotation]”

where the recommended archetype is determined by cycling through the five subjective question styles: $(\text{question_count}) \bmod 5$ selects from the archetypes listed above.

Example. After 3 similarity failures on the 12th question ($s = 3, k = 12$):

DIVERSITY FOCUS: Previous 3 uniqueness attempts suggest trying a new approach

RECOMMENDED STYLE: Focus on “Ask to compare or contrast two related terms or concepts found in the text” (archetype #4, since $12 \bmod 5 = 2$)

This lightweight directive allows the model flexibility while nudging it toward unexplored question patterns. The system does not explicitly list patterns to avoid or aspects to explore, trusting the model’s capability to interpret the diversity signal in conjunction with the semantic similarity filter that validates sufficient variation.

3.5 Complete Prompt Construction Example

The following demonstrates a fully assembled prompt for generating an objective calculation question for multi-hop reasoning testing after 2 similarity failures:

[**BASE TEMPLATE**] *ROLE:* AI Benchmark Scientist | *CONTEXT:* [Technical documentation]
[**TEST-TYPE**] *Multi-Hop Reasoning* | *FOCUS:* Cross-section synthesis | *TRAP:* Constants in section A, voltage in section B, factors in section C
[**FORMAT**] *Mathematical calculation* | [Calculation] Question text | Extract ALL parameters, context-specific formulas, step-by-step with units
[**DIVERSITY (s=2)**] Previous 2 attempts suggest new approach | Explore different calculation scenarios
[**OUTPUT**] JSON only | question_format: "OBJECTIVE" | ground_truth_answer: "The correct answer is [X]. Reasoning: [steps]"

3.6 Prompt Component Interaction

Different prompt components activate dynamically based on generation context:

Base Template (Always): Provides role definition and output format specification, establishing the foundation for all prompts.

Test-Type Instructions (Scenario-dependent): Injects focus area and cognitive demands specific to robustness, multi-hop, or generation quality testing.

Format Specification (Balance tracking): Determines objective vs. subjective question format based on current distribution to ensure coverage.

Diversity Directives ($s > 2$): Activates when similarity failures exceed threshold, providing concise diversity guidance through archetype recommendation.

Archetype Rotation (Subjective only): Cycles through five question styles to maintain stylistic diversity in subjective questions.

This modular design enables fine-grained control over generation characteristics while maintaining consistency through the base template structure.

4 Evaluation Rubrics

Our evaluation framework employs six core dimensions with adaptive rubrics that distinguish between objective and subjective question types. Each dimension uses a 0-10 scoring scale with detailed rubrics, internally normalized to 0-1 by the G-Eval framework. All metrics utilize GPT-4.1-mini as the judge through DeepEval’s G-Eval implementation with chain-of-thought reasoning.

4.1 Completeness

4.1.1 Objective Questions

Evaluation Criteria: Assess completeness of the response including final answer, explanation quality, and reasoning steps.

Evaluation Steps:

1. Check if clear answer is provided
2. Verify calculation methodology and steps
3. Assess relevant formulas or evidence
4. Evaluate reasoning explanations
5. Determine overall completeness

Rubric:

Score	Expected Outcome
9–10	Perfect: Correct answer with complete explanation and clear reasoning
7–8	High: Correct answer with good explanation, minor gaps
5–6	Good: Answer with basic explanation, some missing details
3–4	Moderate: Partial answer or incomplete explanation
1–2	Limited: Minimal answer, lacks proper explanation
0	Poor: No clear answer or completely off-topic

Table 1: Completeness rubric for objective questions

4.1.2 Subjective Questions

Evaluation Criteria: Assess completeness in addressing all key aspects thoroughly.

Evaluation Steps:

1. Identify all key aspects in the question
2. Determine coverage level for each aspect
3. Calculate coverage ratio
4. Consider context adequacy
5. Evaluate overall comprehensiveness

Rubric:

4.2 Depth

4.2.1 Objective Questions

Evaluation Criteria: Assess depth of analysis including sophistication, reasoning, and understanding of principles.

Score	Expected Outcome
9–10	Exceptional: Addresses all key aspects thoroughly
7–8	High: Covers all major aspects clearly with good detail
5–6	Good: Addresses most important aspects reasonably
3–4	Moderate: Covers key aspects with varying depth
1–2	Limited: Addresses some but misses important points
0	Poor: Fails to address key aspects or off-topic

Table 2: Completeness rubric for subjective questions

Evaluation Steps:

1. Evaluate sophistication of approach
2. Assess clarity of step-by-step reasoning
3. Check demonstration of underlying principles
4. Evaluate technical precision
5. Assess overall intellectual rigor

Rubric:

Score	Expected Outcome
9–10	Exceptional: Sophisticated understanding with detailed methodology
7–8	High: Strong analytical thinking with clear solution steps
5–6	Good: Reasonable technical detail with some analysis
3–4	Moderate: Basic approach but relatively shallow analysis
1–2	Limited: Minimal analytical content, superficial approach
0	Poor: No meaningful analysis demonstrated

Table 3: Depth rubric for objective questions

4.2.2 Subjective Questions

Evaluation Criteria: Assess intellectual depth including reasoning sophistication, nuance, and insight quality.

Evaluation Steps:

1. Evaluate reasoning sophistication
2. Assess depth of understanding
3. Check for nuanced perspectives
4. Evaluate insight quality
5. Assess overall intellectual rigor

Rubric:

4.3 Factuality

4.3.1 With Retrieval Context

Evaluation Criteria: Assess factual accuracy against retrieval context, emphasizing verifiable claims and absence of hallucinations.

Evaluation Steps:

Score	Expected Outcome
9–10	Exceptional: Sophisticated understanding with nuanced analysis
7–8	High: Strong analytical thinking with good conceptual depth
5–6	Good: Reasonable analysis with adequate technical detail
3–4	Moderate: Surface-level analysis with limited sophistication
1–2	Limited: Superficial treatment with minimal content
0	Poor: No meaningful analysis or understanding

Table 4: Depth rubric for subjective questions

1. Identify and categorize all factual claims
2. Check accuracy of numbers, dates, specifications
3. Assess contradiction severity
4. Check for hallucinated details
5. Calculate overall accuracy ratio

Rubric:

Score	Expected Outcome
9–10	Perfect: All claims match context exactly, no contradictions
7–8	High: Minor imprecision in non-critical details, major facts correct
5–6	Good: Generally accurate with minor factual gaps
4	Moderate: Noticeable inaccuracies but core info correct
2–3	Limited: Several factual errors or contradictions
0–1	Poor: Major errors or predominantly false information

Table 5: Factuality rubric with retrieval context

4.3.2 Without Retrieval Context

Evaluation Criteria: Assess factual accuracy against expected output, checking for contradictions.

Evaluation Steps:

1. Compare claims with expected output
2. Identify implausible information
3. Assess technical claim consistency
4. Evaluate claim plausibility

Rubric:

Score	Expected Outcome
8–10	High: Claims match expected output, no major contradictions
6–7	Good: Most claims consistent, no significant errors
4–5	Moderate: Some inconsistencies but generally reasonable
2–3	Limited: Notable contradictions or implausible information
0–1	Poor: Major contradictions or obviously incorrect

Table 6: Factuality rubric without retrieval context

4.4 Relevance

4.4.1 Objective Questions

Evaluation Criteria: Assess how precisely response addresses question format requirements.

Evaluation Steps:

1. Check required answer format compliance
2. Evaluate explanation relevance
3. Assess avoidance of tangential info
4. Check focus consistency
5. Evaluate specificity to requirements

Rubric:

Score	Expected Outcome
9–10	Perfect: Directly addresses specific question format
7–8	High: Stays focused with minimal deviation
5–6	Good: Generally addresses question, some unnecessary details
3–4	Moderate: Partially addresses with irrelevant information
1–2	Limited: Somewhat related with significant drift
0	Poor: Fails to address specific requirements

Table 7: Relevance rubric for objective questions

4.4.2 Subjective Questions

Evaluation Criteria: Assess how precisely response stays focused on the specific question.

Evaluation Steps:

1. Analyze each section for relevance
2. Evaluate examples for illustrative value
3. Assess information priority
4. Check focus consistency
5. Calculate relevance ratio

Rubric:

Score	Expected Outcome
9–10	Perfect: Every sentence directly addresses the question
7–8	High: Strongly focused with minimal deviation
5–6	Good: Mostly focused with minor tangential elements
3–4	Moderate: Generally on-topic but includes unnecessary info
1–2	Limited: Partially addresses with significant drift
0	Poor: Minimal connection to actual question

Table 8: Relevance rubric for subjective questions

4.5 Context Utilization

4.5.1 With Retrieval Context

Evaluation Criteria: Evaluate how effectively response integrates retrieved context, comparing golden sources with retrieved sources.

Evaluation Steps:

1. Compare golden vs. retrieved sources
2. Assess retrieval quality
3. Check source attribution accuracy
4. Assess context synthesis quality

Rubric:

Score	Expected Outcome
9–10	Masterful: Seamlessly weaves multiple sources together
7–8	Strong: Effectively uses most relevant context
5–6	Good: Incorporates key context adequately
3–4	Moderate: Uses some context, integration could improve
1–2	Limited: Minimal effective use of available context
0	Poor: Completely ignores or contradicts context

Table 9: Context utilization with retrieval context

4.5.2 Manual Setup Frameworks

Evaluation Criteria: Evaluate evidence of domain-specific knowledge utilization for manual setup frameworks.

Evaluation Steps:

1. Count domain-specific technical terms
2. Identify specific standards mentioned
3. Assess technical depth
4. Check for knowledge indicators

Rubric:

Score	Expected Outcome
8–10	Exceptional: Sophisticated domain expertise with advanced terminology
6–7	Strong: Clear evidence of domain-specific knowledge
4–5	Moderate: Some evidence of specialized knowledge
2–3	Limited: Minimal evidence of specialized knowledge
0–1	Poor: Generic response with no domain-specific content

Table 10: Context utilization for manual setup frameworks

4.6 Support Quality

4.6.1 Objective Questions

Evaluation Criteria: Assess quality of supporting evidence including formulas, steps, and reasoning.

Evaluation Steps:

1. Evaluate formula/equation quality
2. Assess clarity of steps shown
3. Check intermediate results support
4. Evaluate reasoning explanation
5. Assess overall coherence

Rubric:

Score	Expected Outcome
9–10	Outstanding: Exceptional evidence with specific calculations/formulas
7–8	High quality: Strong evidence with good specificity
5–6	Good: Adequate details with reasonable explanation
3–4	Moderate: Some details but could be more specific
1–2	Limited: Minimal evidence or poorly explained
0	Poor: Lacks evidence or contains misleading explanations

Table 11: Support quality for objective questions

4.6.2 Subjective Questions

Evaluation Criteria: Assess quality of supporting evidence including examples and details.

Evaluation Steps:

1. Count and categorize supporting elements
2. Assess example relevance
3. Evaluate technical detail precision
4. Check evidence strength
5. Assess practical value

Rubric:

Score	Expected Outcome
9–10	Outstanding: Exceptional evidence with specific, concrete details
7–8	High quality: Strong evidence with good specificity
5–6	Good: Adequate details with reasonable examples
3–4	Moderate: Some details but could be more specific
1–2	Limited: Minimal evidence or poorly chosen examples
0	Poor: Lacks evidence or contains misleading examples

Table 12: Support quality for subjective questions

4.7 Implementation Details

All metrics are implemented using DeepEval’s G-Eval framework with the following configuration:

- **Model:** GPT-4.1-mini
- **Scoring:** Rubric-based evaluation with 0–10 scale, internally normalized to 0–1
- **Reasoning:** Chain-of-thought prompting with explicit evaluation steps
- **Rate Limiting:** 0.5 second intervals between evaluations
- **Output Format:** Each evaluation returns (1) normalized score, (2) binary success indicator, and (3) natural language reasoning

The adaptive evaluation approach distinguishes between objective questions (requiring specific answers) and subjective questions (requiring open-ended explanations), applying tailored rubrics and evaluation steps for each question type.

5 Evaluation Platform Implementation

This appendix provides implementation details for the evaluation platform described in Section ??, based on the actual adapter implementations.

5.1 Framework Capability Matrix

Table 13 summarizes the capabilities of each evaluated RAG framework based on their API implementations.

Table 13: RAG Framework Capabilities

Framework	API Upload	Sources Exposed	Model Config	Manual Setup
AnythingLLM	✓	✓	✓	Recommended
RAGFlow	✓	✓	✓	Recommended
MaxKB	—	—	✓	Required
Metaso	✓	Partial	—	Required

Key Differences. **AnythingLLM** provides full retrieval transparency through its sources field, returning document chunks with relevance scores. The adapter extracts sources through multiple fallback methods to handle response format variations.

RAGFlow uses an OpenAI-compatible API endpoint (`/api/v1/chats_openai/{chat_id}/chat/completions`), returning responses in standard OpenAI format. Source attribution requires inference from response content since the compatibility layer abstracts retrieval details.

MaxKB does not expose retrieval context through its API by design—sources are visible only in the web interface. Our adapter implements heuristic detection using domain-specific indicators:

```
knowledge_indicators = ["according to",
    "based on", "standard specifies",
    "SEMI", "specification",
    "technical documentation"]
professional_terms = ["SEMI", "E1", "E30",
    "wafer", "process", "GEM",
    "SECS", "HSMS"]
```

Metaso returns structured references including title, author, date, and external links, but uses a proprietary DeepSeek-based model that cannot be reconfigured, limiting its participation to cross-framework (not cross-model) evaluation.

5.2 Adaptive Evaluation Strategy

The evaluation layer (`evaluator_gpt.py`) adapts metric computation based on framework capabilities defined in `FRAMEWORK_CAPABILITIES`:

```
FRAMEWORK_CAPABILITIES = {
    'anythingllm': {
        'supports_api_upload': True,
        'requires_manual_setup': True,
        'provides_sources': True
    },
    'ragflow': {
        'supports_api_upload': True,
        'requires_manual_setup': True,
        'provides_sources': True
    },
    'maxkb': {
        'supports_api_upload': False,
        'requires_manual_setup': True,
        'provides_sources': False
    },
    'metaso': {
        'supports_api_upload': True,
        'requires_manual_setup': True,
        'provides_sources': True
    }
}
```

```

'maxkb': {
  'supports_api_upload': False,
  'requires_manual_setup': True,
  'provides_sources': False
},
...
}

```

For frameworks with `provides_sources: False`, the Context Utilization metric switches from direct source comparison to heuristic inference based on response characteristics. This ensures fair evaluation across frameworks with different transparency levels.

5.3 Dual-Mode Implementation

The dual-mode assessment protocol is implemented through the test orchestrator (`test_rag_frameworks_v2.py`):

- **Mode A (with_kb):** Standard RAG query through framework’s native retrieval pipeline
- **Mode B (without_kb):** Gold context injected into prompt, bypassing retrieval to isolate generation capabilities

Cross-mode score differences ($\Delta = \text{Mode B} - \text{Mode A}$) enable failure attribution: large positive Δ indicates retrieval failures; consistently low scores in both modes indicate generation-stage weaknesses.

5.4 Output Format

Evaluation results are stored in structured JSON with per-metric scores, reasoning explanations, and metadata:

```

{
  "test_id": "robustness_001",
  "framework": "anythingllm",
  "question_format": "OBJECTIVE",
  "metric": "factuality",
  "score": 0.85,
  "reason": "Response accurately...",
  "has_retrieval_context": true,
  "requires_manual_setup": true
}

```

The `question_format` field enables metric-specific rubric selection, applying calculation-focused criteria for objective questions and comprehensiveness-focused criteria for subjective questions as detailed in Appendix 4.

6 Complete Experimental Results

This appendix provides the complete six-dimensional evaluation results for all model-configuration combinations.

Table 14: DeepSeek-v3.2-Exp performance across context configurations.

Config	Fact.	Depth	Comp.	Rel.	Ctx.U.	Supp.	Avg.
(4K, 1K)	0.626	0.554	0.592	0.627	0.783	0.534	0.619
(8K, 2K)	0.632	0.602	0.640	0.661	0.804	0.594	0.656
(10K, 4K)	0.663	0.606	0.651	0.669	0.814	0.609	0.669
(12K, 4K)	0.690	0.640	0.688	0.701	0.824	0.644	0.698
(14K, 4K)	0.744	0.693	0.752	0.740	0.845	0.709	0.747
(16K, 4K)	0.792	0.718	0.783	0.781	0.868	0.735	0.780
(18K, 4K)	0.837	0.755	0.820	0.813	0.884	0.776	0.814
(20K, 4K)	0.865	0.796	0.860	0.850	0.901	0.823	0.849
(24K, 4K)	0.868	0.812	0.880	0.857	0.915	0.839	0.862
(28K, 4K)	0.892	0.811	0.879	0.868	0.916	0.842	0.868
(32K, 4K)	0.904	0.838	0.898	0.875	0.912	0.872	0.883

Table 15: Qwen-Plus performance across context configurations.

Config	Fact.	Depth	Comp.	Rel.	Ctx.U.	Supp.	Avg.
(4K, 1K)	0.663	0.662	0.686	0.677	0.805	0.641	0.689
(8K, 2K)	0.665	0.710	0.724	0.704	0.801	0.696	0.717
(10K, 4K)	0.680	0.736	0.764	0.736	0.828	0.724	0.745
(12K, 4K)	0.697	0.749	0.765	0.722	0.823	0.747	0.751
(14K, 4K)	0.723	0.772	0.804	0.760	0.834	0.767	0.777
(16K, 4K)	0.750	0.789	0.809	0.768	0.848	0.770	0.789
(18K, 4K)	0.732	0.783	0.813	0.774	0.844	0.783	0.788
(20K, 4K)	0.778	0.810	0.830	0.798	0.852	0.801	0.812
(24K, 4K)	0.833	0.829	0.861	0.827	0.882	0.835	0.845
(28K, 4K)	0.866	0.849	0.894	0.859	0.916	0.857	0.874
(32K, 4K)	0.855	0.829	0.867	0.847	0.889	0.831	0.853

Table 16: Gemini-2.5-Flash performance across context configurations.

Config	Fact.	Depth	Comp.	Rel.	Ctx.U.	Supp.	Avg.
(4K, 1K)	0.600	0.374	0.399	0.449	0.669	0.350	0.474
(8K, 2K)	0.643	0.536	0.563	0.583	0.757	0.523	0.601
(10K, 2K)	0.666	0.550	0.595	0.622	0.758	0.622	0.636
(12K, 4K)	0.701	0.641	0.692	0.708	0.819	0.651	0.702
(14K, 4K)	0.743	0.655	0.709	0.726	0.839	0.656	0.721
(16K, 4K)	0.760	0.669	0.733	0.731	0.816	0.676	0.731
(18K, 4K)	0.776	0.710	0.771	0.784	0.847	0.728	0.769
(20K, 4K)	0.833	0.738	0.804	0.808	0.878	0.752	0.802
(24K, 4K)	0.854	0.765	0.827	0.837	0.882	0.790	0.826
(28K, 4K)	0.901	0.818	0.886	0.890	0.907	0.852	0.876
(32K, 4K)	0.869	0.775	0.844	0.836	0.889	0.801	0.836

Table 17: Qwen-2.5-72B-Instruct performance with extended context windows.

Config	Fact.	Depth	Comp.	Rel.	Ctx.U.	Supp.	Avg.
(4K, 1K)	0.511	0.568	0.603	0.589	0.758	0.536	0.594
(8K, 2K)	0.544	0.590	0.629	0.605	0.785	0.560	0.619
(16K, 4K)	0.643	0.661	0.713	0.692	0.798	0.630	0.690
(32K, 4K)	0.794	0.762	0.827	0.812	0.852	0.764	0.802
(64K, 6K)	0.801	0.759	0.820	0.796	0.856	0.738	0.795
(128K, 8K)	0.797	0.766	0.829	0.812	0.863	0.762	0.805

Table 18: Cross-model comparison at key context configurations.

Model	4K	16K	28K	32K
DeepSeek-v3.2-Exp	0.619	0.780	0.868	0.883
Qwen-Plus	0.689	0.789	0.874	0.853
Gemini-2.5-Flash	0.474	0.731	0.876	0.836
Qwen-2.5-72B	0.594	0.690	—	0.802

Table ?? highlights the three scaling behaviors discussed in Section ??:

- **Logarithmic Growth (DeepSeek):** Consistent improvement from 4K (0.619) to 32K (0.883) with no saturation or decline.
- **Early Strength, Late Saturation (Qwen-Plus):** Highest at 4K (0.689), peaks at 28K (0.874), declines at 32K (0.853).
- **Cold-Start Dynamics (Gemini):** Lowest at 4K (0.474), peaks at 28K (0.876), declines at 32K (0.836).

The 28K→32K decline observed in Qwen-Plus and Gemini suggests that these architectures experience information overload at extreme context lengths, while DeepSeek’s continued improvement indicates superior noise tolerance mechanisms.